AK VIT Vellore

AK

REG. NO: 26BYB0282 · B.TECH CSE · VIT VELLORE

Ayan Kaushik

SOFTWARE DEVELOPER · SYSTEMS ENTHUSIAST · PROBLEM SOLVER

"Committed to engineering dependable systems, robust command-line security tools, and intelligent software applications through first-principles thinking and technical precision."

Explore Projects | View Resume / CV | Contact Information

About Me

AK VIT Vellore

Profile & Professional Vision

I am **Ayan Kaushik**, an undergraduate student pursuing a Bachelor of Technology in **Computer Science and Engineering** at **Vellore Institute of Technology, Vellore**. My academic and project pursuits center around the convergence of low-level systems programming, secure developer utilities, and artificial intelligence.

> *"I believe high-quality engineering is rooted in understanding how software interacts with system memory, kernels, and developer workflows."*

Technical Interests: I am deeply fascinated by operating system architecture, POSIX terminal environments, heuristic command defense, and building intelligent tools that streamline human-computer interaction.

Core Strengths: Structured analytical reasoning, algorithmic decomposition in C++, rapid modular prototyping in Python, and articulate technical documentation.

Career Direction: My professional goal is to contribute to pioneering systems software engineering and software reliability, crafting production-grade software that is secure, computationally efficient, and user-centric.

Vellore Institute of Technology, Vellore, Tamil Nadu

Seeking Research & Internship Opportunities

>_ Systems & CLI

Exploration of Unix primitives, defensive shell scripting, and memory management.

Artificial Intelligence

Applied machine learning workflows for focus automation, NLP, and cognitive tooling.

Methodology

Rigorous test-driven engineering, clean modular structure, and clear documentation.

Target Roles

Systems Software Engineer, Backend/Core Developer, AI Applications Engineer.

Education

CURRENT PROGRAMME · UNDERGRADUATE

Reg. No: 26BYB0282

Vellore Institute of Technology (VIT), Vellore

Bachelor of Technology (B.Tech) — Computer Science and Engineering

Pursuing a rigorous curriculum emphasizing theoretical computer science, systems design, software engineering practices, and technical communication. Actively involved in project-based learning and peer technical collaborations.

KEY COURSEWORK & ACADEMIC FOCUS:

- Data Structures & Algorithms
- Object-Oriented Programming (C++)
- Operating Systems & Architecture
- Technical English Communication (BAENG101P)
- Discrete Mathematics & Logic
- Database Management Systems

Demonstrated consistent commitment to technical excellence and laboratory problem solving.

HIGHER SECONDARY EDUCATION (CLASS XII) Score: 90.2%

Senior Secondary STEM & Advanced Sciences

Physics, Chemistry, and Mathematics (PCM) with Computer Science · **90.2% Aggregate**

Built strong foundational competence in calculus, differential equations, classical physics, and logical abstraction, establishing a disciplined analytical mindset for engineering studies.

`Score: 90.2%` `Advanced Mathematics` `Scientific Method` `Algorithmic Thinking`

SECONDARY SCHOOL CERTIFICATE (CLASS X) Score: 94.6%

Secondary Education & Foundational Sciences

Core Sciences, Mathematics, English, and Social Sciences · **94.6% Aggregate**

Demonstrated academic distinction across all disciplines with strong performance in mathematics and sciences, cultivating early aptitude for scientific inquiry and technology.

`Score: 94.6%` `Mathematical Foundations` `Academic Distinction`

Skills & Competencies

A structured breakdown of technical proficiency, digital toolchain mastery, communication capabilities, and professional workplace attributes.

Technical Skills

Core languages and low-level development paradigms.

Python | C++ (Modern) | Bash / Shell | C | SQL | Data Structures

Emphasis on memory safety, algorithmic complexity & POSIX compliance.

Digital Tools

Modern developer infrastructure and platforms.

Linux / Unix CLI | Git & GitHub | VS Code | GDB Debugger | Docker Basics | Markdown

Version control, automated build environments & issue tracking.

Communication

Clarity in verbal, written & technical discourse.

Technical Writing | Presentations | Documentation | Active Listening | Peer Review

Clear articulation of technical ideas to cross-disciplinary audiences.

Professional

Workplace efficacy, ethics & teamwork.

Problem Solving | Time Management | Collaboration | Critical Thinking | Adaptability

Constructive team participation and goal-oriented project execution.

Projects & Evidence

Demonstrated engineering work showcasing practical application of computer science principles, system safety, and artificial intelligence solutions.

AK VIT Vellore

Refocus AI

Intelligent Focus & Deep-Work Engine

Live Project

PROBLEM STATEMENT:

Students and engineers lose up to 40% of productive time to frequent cognitive interruptions and context switching during computer sessions.

ENGINEERING SOLUTION:

Engineered an intelligent, context-aware productivity assistant in Python that monitors distraction triggers, schedules automated deep-work intervals, and generates diagnostic session analytics.

- Zero-latency local background daemon using lightweight Python threading
- Heuristic telemetry to quantify focus metrics and attention drift
- Modular architecture with clean separation of logic, analytics, and UI alerts

Python | AI / Machine Learning | Telemetry | FastAPI | Productivity

Repository Evidence →

Public Codebase

AK VIT Vellore

BashGuard

Terminal Security & Command Interceptor

Security Tool

PROBLEM STATEMENT:

Terminal operators and developers risk catastrophic filesystem corruption and credential leakage when executing unvetted scripts or malformed recursive shell commands.

ENGINEERING SOLUTION:

Developed a defensive CLI guard in C++ and POSIX Shell that intercepts shell inputs before kernel execution, performing pattern recognition and blocking destructive commands.

- Pre-execution shell hooks intercepting destructive operations (e.g., recursive root deletes)
- Heuristic static detection for untrusted pipe-to-bash downloads
- Sub-millisecond latency footprint engineered in native C++

C++ | Bash Scripting | Linux / POSIX | Defensive Security | CLI Architecture

Repository Evidence →

Security Suite

Academic Excellence

Consistently high performance in STEM and computer science coursework at VIT Vellore, with rigorous participation in programming laboratories.

Hackathons & Problem Solving

Active participant in competitive programming events and student hackathons focusing on AI agent design and software automation.

AK VIT Vellore

Technical Documentation

Authored clear Markdown READMEs, operational instructions, and architectural specifications across all personal software repositories.

Resume / Curriculum Vitae

Complete summary of qualifications, education, technical competencies, and project records. You can inspect the interactive resume card below or trigger a formatted print/save.

AK VIT Vellore

EMAIL
ayanrunning99@gmail.com

INSTITUTION
VIT Vellore, Tamil Nadu, India

PORTFOLIO & CODE
cypher07-ai.github.io/portfolio

CAREER OBJECTIVE & PROFILE SUMMARY

Aspiring Software Engineer with a solid foundation in Computer Science principles, C++, Python, and Linux systems architecture. Dedicated to building reliable, high-performance applications and defensive developer utilities. Seeking engineering internships and collaborative research opportunities to apply algorithmic problem-solving to real-world software.

ACADEMIC QUALIFICATIONS

Vellore Institute of Technology (VIT), Vellore — 2025 – Present
Bachelor of Technology in Computer Science and Engineering (B.Tech CSE) · Reg. No: 26BYB0282

Higher Secondary Education (Class XII) — 90.2%
Physics, Chemistry, Mathematics (PCM) with Computer Science · Aggregate Score: **90.2%**

Secondary School Certificate (Class X) — 94.6%
Core Sciences, Mathematics, English, Social Sciences · Aggregate Score: **94.6%**

KEY TECHNICAL PROJECTS

Refocus AI — Intelligent Focus Engine — Python, Machine Learning, FastAPI

- Designed context-aware productivity daemon to detect workflow interruptions and optimize deep work sessions.
- Implemented modular background telemetry tracking user attention drift with near-zero resource utilization.

BashGuard — Terminal Execution Guard — C++, Bash, POSIX, Linux

- Developed pre-execution interception hooks in native C++ to prevent accidental recursive directory wipeouts.
- Created heuristic static analysis engine checking pipe-to-shell patterns for untrusted network payloads.

TECHNICAL & PROFESSIONAL COMPETENCIES

Languages & Core:

Tools & Platforms:

Python, C++, C, Bash/POSIX Scripting, SQL, Data Structures & Algorithms

Linux CLI, Git, GitHub, VS Code, GDB Debugger, Technical Documentation

Contact & Professional Links

Feel free to reach out for academic discussions, collaborative technical projects, or internship inquiries.

DIRECT EMAIL

ayanrunning99@gmail.com

Copy Email to Clipboard

GITHUB PROFILE

@cypher07-ai

Source code for Refocus AI, BashGuard, and academic lab experiments.

Visit GitHub Profile →

ACADEMIC LOCATION

Vellore Institute of Technology

Vellore – 632014, Tamil Nadu, India.

School of Computer Science & Engineering

Let's Connect and Collaborate

Whether you have feedback on my projects, an internship opportunity, or wish to discuss computer science concepts, my inbox is open.

Compose an Email